

Empirical Research in Finance

4. Dummy variables

Beatriz Mariano

Outline

Dummies as regressors

- Intercepts

- Intercept interactions

- Slope interactions

Dummies as dependent variables

- Linear probability model

Dummies as regressors

Dummy variables (binary variables) take only two values: 0 or 1.

We use them to encode *qualitative* information:

- ▶ bank vs non-bank
- ▶ bankrupt vs non-bankrupt
- ▶ crisis vs non-crisis

With K categories use $K - 1$ dummies to avoid perfect multicollinearity. Coefficients are read *relative to the omitted base category*.

When to convert ordinal or quantitative info to dummies

We can also convert ordinal or quantitative information into dummies:

- ▶ Ratings: AAA, AA, A, BBB, ...
coded as a set of dummies (1/0) instead of a single 1,2,3,4 score
- ▶ Size: large (size > median) vs. small
instead of size in millions

Why bother?

Reduces complexity, makes results easier to communicate, and lets the data tell us whether the effect is *linear* in the underlying scale – which is rarely true for ratings or size.

Finance example: bond ratings and issuance yield

$$\text{iyield} = \beta_0 + \beta_1 \text{rating} + \dots \quad \text{rating} \in \{1, 2, 3, 4, \dots\}$$

Questions

On the rating scale 1 = AAA, 2 = AA, 3 = A, 4 = BBB:

- ▶ Do you expect β_1 to be *positive* or *negative*?
- ▶ Is the difference in yield between AAA and AA likely to be the same as between A and BBB?

Better-specified model – one dummy per rating bucket:

$$\text{iyield} = \beta_0 + \beta_2 \text{AA} + \beta_3 \text{A} + \beta_4 \text{BBB} + \dots$$

Omitted (base) category: AAA $\implies \beta_4$ is the *average* difference in yield between a BBB-bond and an AAA-bond, holding everything else fixed.

Bond ratings – estimated output (illustrative)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.213	0.082	51.4	<2e-16 ***
AA	0.184	0.094	1.96	0.050 *
A	0.612	0.101	6.06	2.1e-09 ***
BBB	1.557	0.118	13.20	<2e-16 ***
...				

- ▶ AAA is the base $\Rightarrow$ each coefficient is “yield premium vs. AAA, all else equal”.
- ▶ The yield jump AAA $\rightarrow$ A is much larger than AA $\rightarrow$ AAA $\Rightarrow$ the effect is **not linear** in the rating score.

Dummies as regressors – Intercepts

Question: do women earn less than men with the same education, experience and tenure?

Wooldridge Example 7.5 – WAGE1 dataset

- ▶ wage: average hourly earnings in USD
- ▶ female: =1 if female
- ▶ married: =1 if married
- ▶ educ, exper, tenure: years of schooling / experience / tenure

$$\log \text{wage}_i = \beta_0 + \delta_0 \text{female}_i + \beta_1 \text{educ}_i + \cdots + u_i$$

```
> require(wooldridge); attach(wage1)
> lwage <- log(wage); expersq <- exper^2; tenuresq <- tenure^2
> m <- lm(lwage ~ female + educ + exper + expersq
+          + tenure + tenuresq)
```

Reading the output (only the rows that matter)

Coefficients (relevant rows):

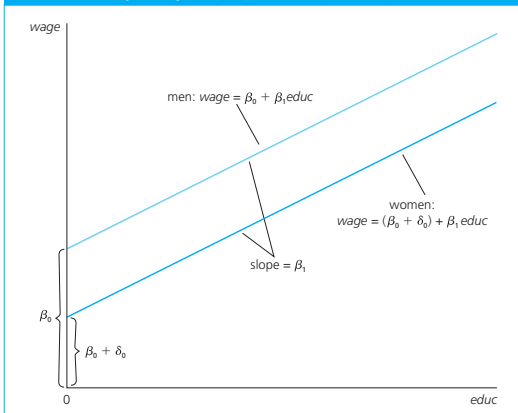
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.41669	0.09893	4.21	3.0e-05	***
female	-0.29651	0.03581	-8.28	1.0e-15	***
educ	0.08020	0.00676	11.87	< 2e-16	***
...					

Adjusted R-squared: 0.434

- ▶ **Women earn $\approx 29.7\%$ less** than men, holding educ, exper and tenure fixed.

Visualizing

FIGURE 7.1 Graph of $wage = \beta_0 + \delta_0 \text{ female} + \beta_1 \text{ educ}$ for $\delta_0 < 0$.



Source: Wooldridge (graph for wage, not log wage).

Adding a second dummy: married

```
> m <- lm(lwage ~ female + married + educ + exper + expersq  
+          + tenure + tenuresq)
```

	Estimate	Std. Error	t value	Pr(> t)
female	-0.29018	0.03611	-8.04	6.3e-15 ***
married	0.05292	0.04076	1.30	0.195
educ	0.07915	0.00680	11.64	< 2e-16 ***
...				

Marriage on its own does not seem to matter – coefficient is small and insignificant. But this is misleading: the *effect of marriage* may differ between men and women. Next: interaction.

Intercept interactions – the 2×2 logic

Does the effect of marriage depend on whether a person is male or female?

	unmarried	married
male	base	+ married
female	+ female	+ female + married + female:married

Include the constituent dummies on their own. An interaction without its main effects is almost never what you want.

Intercept interactions – the regression

```
> m <- lm(lwage ~ female*married + educ + exper + expersq  
+           + tenure + tenuresq)
```

In R, `female*married` is shorthand for `female + married + female:married` – it expands to all three terms automatically.

Intercept interactions – estimated output

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.32138	0.10001	3.21	0.00139	**
female	-0.11035	0.05574	-1.98	0.04827	*
married	0.21268	0.05536	3.84	0.00014	***
female:married	-0.30059	0.07177	-4.19	3.3e-05	***
educ	0.07891	0.00669	11.79	< 2e-16	***
...					

Decomposition by group

Estimated coefficients (from previous slide):

const	female	married	female:married
+0.321	-0.110	+0.213	-0.301

	const	female	married	female_married	total
single men	+0.321				= 0.321
married men	+0.321		+0.213		= 0.534
single women	+0.321	-0.110			= 0.211
married women	+0.321	-0.110	+0.213	-0.301	= 0.123

Numbers are predicted **log wage** holding all other controls at zero. The *base / omitted category* is single men.

What the decomposition tells us

- M: A married man earns on average **+21.3%** more than a single man. (0.213)
- F: A married woman earns on average **-8.8%** less than a single woman. $(0.123 - 0.211 = -0.088)$
- Δ : Difference of **-30.1%** between the two “marriage premia” is significant.
 $(-0.088 - 0.213 = -0.301)$

Dummies as regressors – Slope interactions

We can also interact a dummy with a *continuous* variable.

Another question: does the relation between firm size and CEO pay differ for long-tenured CEOs?

Wooldridge ceosal2 dataset (177 US firms)

- ▶ lsalary: log of CEO total compensation
- ▶ lsales: log of firm sales (size)
- ▶ ceoten: years as CEO
- ▶ HighTenure: = 1 if $\text{ceoten} \geq \overline{\text{ceoten}}$ (≈ 7.95 y)

```
> ceosal2$HighTenure <- as.numeric(ceosal2$ceoten >=
+                               mean(ceosal2$ceoten))
> m <- lm(lsalary ~ lsales*HighTenure + ceoten,
+         data = ceosal2)
```

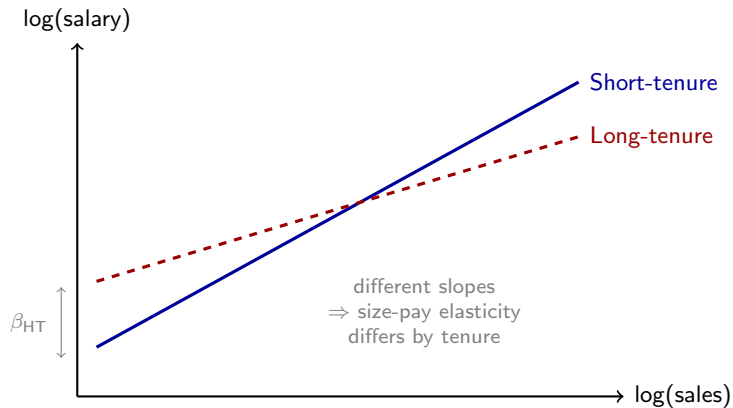
Estimated output

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	4.7753	0.2531	18.87	<2e-16	***
lsales	0.2374	0.0338	7.02	5.0e-11	***
HighTenure	0.4524	0.4121	1.10	0.274	
lsales:HighTenure	-0.0225	0.0547	-0.41	0.681	
ceoten	-0.0039	0.0082	-0.47	0.637	

N = 177	R ² = 0.323				

- ▶ Estimated size-pay elasticity:
 - ▶ short-tenure CEOs: 0.237
 - ▶ long-tenure CEOs: $0.237 - 0.022 = 0.215$
- ▶ Long-tenure CEOs appear to have a slightly *weaker* size-pay link, but the -0.022 gap is not significant on its own.

Visualising the slope interaction



HighTenure shifts the *intercept*; `lsales:HighTenure` shifts the *slope*.

So...tenure doesn't matter for CEO pay?

Look at the two coefficients that involve HighTenure:

HighTenure	0.452	(n.s.)
lsales:HighTenure	-0.022	(n.s.)

The puzzle

Both look insignificant on their own. Does that mean tenure is not associated with CEO pay – neither in level nor in the size-pay slope?

No – the two coefficients are highly *correlated* in the sample, so each *t*-statistic is misleadingly small.

The right test is a joint *F*-test:

$$H_0 : \beta_{\text{HighTenure}} = 0 \text{ AND } \beta_{\text{lsales:HighTenure}} = 0$$

The joint F -test

```
> require(car)
> linearHypothesis(m, c('HighTenure = 0',
+                        'lsales:HighTenure = 0'))
```

Linear hypothesis test						
	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	174	45.315				
2	172	43.764	2	1.552	3.05	0.0500 *

$F = 3.05$, $p = 0.05 \implies$ taken jointly, CEO tenure does play a role in the pay equation.

Implementation tips – R, Stata, Python

R (used in this lecture)

Single dummy	<code>lm(y ~ D + x, data = df)</code>
Many categories	<code>lm(y ~ factor(sector) + x, data = df)</code>
Interaction	<code>lm(y ~ x*D, data = df)</code> (<i>includes x, D, x:D</i>)
Joint test	<code>car::linearHypothesis(m, c('D=0', 'x:D=0'))</code>

Stata

Single dummy	<code>reg y D x</code>
Many categories	<code>reg y i.sector x</code>
Interaction (manual)	<code>gen xD = x*D then reg y x D xD</code>
Interaction (factor op)	<code>reg y c.x##i.D</code> (<i>includes x, D, c.x#i.D</i>)
Joint test	<code>test D xD or testparm i.D c.x#i.D</code>

Python (statsmodels)

Single dummy	<code>smf.ols('y ~ D + x', data=df).fit()</code>
Many categories	<code>smf.ols('y ~ C(sector) + x', data=df).fit()</code>
Interaction	<code>smf.ols('y ~ x*D', data=df).fit()</code>
Joint test	<code>m.f_test('D = 0, x:D = 0')</code>

Dummies as dependent variables

Many finance outcomes are binary:

- ▶ $y = 1$ firm has *defaulted*; $y = 0$ otherwise
- ▶ $y = 1$ bank receives *bailout*; $y = 0$ otherwise
- ▶ $y = 1$ loan application *approved*; $y = 0$ denied
- ▶ $y = 1$ firm pays *dividend*; $y = 0$ otherwise

Question: how do we model the *probability* of $y = 1$?

Linear Probability Model (LPM)

Just run OLS on the binary outcome:

$$y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + u_i$$

Then β_j is read as the change in $\Pr(y = 1 \mid x)$ (in *percentage points*) for a one-unit change in x_j .

Why we still teach this

The LPM is simpler than logit/probit, the coefficients are directly interpretable, and you can use everything you already know about OLS. It is the right starting point for almost any binary outcome.

Worked example – top-paid CEOs

Wooldridge dataset `ceosal2` (already used in Section 1):

- ▶ `TopPaid`: 1 if salary $\geq$ sample median ($\approx$ \$707k) – dummy, dependent variable
- ▶ `lsales`: log of firm sales
- ▶ `ceoten`: years as CEO
- ▶ `age`, `college`, `profmarg`: standard controls

```
> attach(ceosal2)
> ceosal2$TopPaid <- as.numeric(ceosal2$salary >=
+                               median(ceosal2$salary))
> m <- lm(TopPaid ~ lsales + ceoten + age + college
+          + profmarg)
```

Note: different from `lwage` regression – where the LHS was continuous (log wage) and slopes read as approximate % changes.

Output and interpretation

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.78061	0.36325	-2.15	0.0330	*
lsales	0.15340	0.02434	6.30	2.4e-09	***
ceoten	0.01231	0.00509	2.42	0.0166	*
age	0.00033	0.00443	0.08	0.9401	
college	0.06045	0.20806	0.29	0.7717	
profmarg	-0.00021	0.00190	-0.11	0.9128	

N = 177	R ² = 0.217				

A 10% larger firm $\Rightarrow$ +1.5 percentage points in the probability of being a top-paid CEO.

LPM issues – predicted probabilities can leave $[0, 1]$

Take the LPM we just estimated and plug numbers in:

$$\begin{aligned}\hat{\Pr}(\text{TopPaid} = 1) = & -0.781 + 0.153 \text{lsales} + 0.012 \text{ceoten} \\ & + 0.0003 \text{age} + 0.060 \text{college} - 0.0002 \text{profmargin}\end{aligned}$$

Hold $\text{ceoten}=10$, $\text{age}=55$, $\text{college}=1$, $\text{profmargin}=10$, and vary lsales :

$$\hat{\Pr}(\text{TopPaid} = 1) = -0.581 + 0.153 \text{lsales}$$

lsales	$\hat{\Pr}(\text{TopPaid} = 1)$	
3	-0.121	< 0 – impossible
5	0.186	ok
7	0.493	ok
9	0.800	ok
11	1.107	> 1 – impossible

Nothing in OLS bounds the predicted probability between 0 and 1. Moreover, R^2 is not so clear to interpret with a binary outcome.

Three things to remember

1. Dummy on the right-hand side?

With K categories, use $K - 1$ **dummies**. Every coefficient is a difference *vs. the base*.

2. Interaction in the model?

Always keep the main effects. If a dummy *and* its interaction both look insignificant $\Rightarrow$ run the **joint F -test** before saying “no effect”.

3. Dummy on the left-hand side (LPM)?

It's still OLS. Read coefficients in **percentage points**, not percent.

*Before interpreting any coefficient, check the **units**, the **base**, and the **right test**.*

Reading

- ▶ Wooldridge, chapters 1–6 (regression refresher).
- ▶ Wooldridge, sections 7.1–7.4 on dummy variables.